MILITARY SPECIFICATION

INSULATION FELT, THERMAL,

GLASS FIBER

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers glass fiber insulation felt for thermal control of machinery and equipment.

1.2 Classification. Insulation felt shall be of the following types, as specified (see 6.2):

Type I - Felted rovings.
Type II - Laminated and felted.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
CCC-T-191 - Textile Test Methods.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock Corner.

MILITARY
MIL-P-116 - Preservation, Methods of.
MIL-Y-1140 - Yarn, Cord, Sleeving; Cloth, and Tape Glass.

STANDARD

MILITARY
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING MATERIALS (ASTM)
C-177 - Thermal Conductivity of Materials by Means of the Guarded Hot Plate, Method of Test for.

(Applications for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, Penna. 19103.)

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Material and construction.

3.1.1 Type I. The material shall consist of staple glass fibers felted into rovings and woven or bound with wire-inserted asbestos thread.

3.1.2 Type II. The material shall consist of 100 percent glass fibers in accordance with MIL-Y-1140, composed of adhered laminates without use of binders. No organic fiber or shot shall be included in the material used in the manufacture of the insulations.

3.2 Dimensions and weight.

3.2.1 Type I.

3.2.1.1 Length. Unless otherwise specified (see 6.2), the insulation shall be furnished in rolls of 25 feet (ft.) in length.

3.2.1.2 Width. Unless otherwise specified (see 6.2), width of each roll shall be 60 inches.

3.2.1.3 Thickness and weights. The insulation shall be furnished in thicknesses shown in table I as specified (see 6.2), and shall vary not more than ± 10 percent from the weight specified for the ordered thickness.

Table I - Thickness and weights, type I.

Nominal thickness	Thickness tolerance ±	Nominal weight per square foot (sq.ft.)
Inches	Inch	Ounces
3/4	1/8	9.0
1	1/8	12.0
1-1/2	1/8	18.0

3.2.2 Type II.

3.2.2.1 Length. Unless otherwise specified (see 6.2), the insulation shall be furnished in rolls of 25 yard (yd.) lengths.

3.2.2.2 Width. Unless otherwise specified (see 6.2), width of roll shall be 60 inches.

3.2.2.3 Thickness and weights. The insulation shall be furnished in the thicknesses shown in table II, as specified (see 6.2) and shall vary not more than ± 10 percent from the weight specified for the ordered thickness.

Table II - Thickness and weights, type II.

Nominal thickness	Thickness tolerance ±	Nominal weight per sq. ft.
Inch	Inch	Ounces
1/2	1/8	5.0
3/4	1/8	10.0
1	1/8	15.0

3.3 Thermal conductivity. Thermal conductivity (k) in British thermal unit (Btu) per square foot per hour of insulation for 1° Fahrenheit (F.) gradient per inch thickness shall not exceed the values at the mean temperatures shown in table III (see 4.4.3).

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Table III - Thermal Conductivity.

Mean temperature	Thermal conductivity (k)	
Degrees F.	Type I	Type II
300	0.45	0.35
500	0.56	0.45
700	0.70	0.60

3.4 Fineness of fiber. The diameter (dia.) of the individual fibers shall average 0.00035 inch or less. Ninety percent of the fibers shall be less than 0.00040 inch dia. (see 4.4.4).

3.4.1 Length of fiber. Fibers less than one inch in length shall not be used in manufacture of the insulation (see 4.4.4.1).

3.5 Alkalinity. The alkalinity of the finished material expressed as sodium oxide (Na_2O) shall not exceed 0.20 percent (see 4.4.5).

3.6 Fusing temperature. The fusing temperature of the fibers shall be not less than 1300°F. (see 4.4.6).

3.7 Stability. The insulation shall reveal no apparent physical change in the glass fibers. The glass fibers shall show no appreciable deceleration when tested in accordance with 4.4.7.

3.8 Tensile strength, (type II only). The tensile strength before and after heating to 1200° F. shall not be less than 5.0 pounds per square inch (psi) (see 4.4.8).

3.9 Workmanship. The workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot. For purposes of sampling, a lot shall consist of not more than 40,000 sq. ft. of glass fiber insulation felt of the same type and thickness.

4.2.2 Sampling for quality conformance inspection.

4.2.2.1 Sampling for examination. A random sample of rolls shall be selected from each lot of material for the examination specified in 4.3.1, with lot acceptance based on table IV.

4.2.2.2 Sampling for tests. From each lot a maximum of six samples, 12 by 60 inches, shall be selected at random for the tests specified in 4.3.2. Each specimen shall be taken from a different roll.

Table IV - Sampling for examination.

Number of rolls in inspection lot	Number of rolls in sample	Acceptance number (defectives)	Rejection number (defectives)
15 and under	5	0	1
16 to 40	7	0	1
41 to 65	10	0	1
66 to 110	15	1	2
Over 110	25	2	3

4.3 Examination and tests.

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4.3.1 Examination. Each of the sample rolls selected in accordance with 4.2.2.1 shall be surface examined, weighed, and measured to determine conformance with the requirements of this specification which do not require tests. Examination shall be conducted as specified in table V, and at least 15 ft. in length of each roll shall be examined. Any roll in the sample containing one or more visual or dimensional defects shall not be offered for delivery, and if the number of defective rolls in any sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by the sample.

Table V - Classification of defects.

Categories	Defects
Critical:	Non defined:
Major:	
101	Type of insulation not as specified.
102	Material less than minimum requirements; evidence of unauthorized materials and processes used; insulation not composed of specified materials.
103	Construction of type I insulation nonconforming or not as specified; not woven or bound with wire-inserted asbestos thread as specified.
104	Construction of type II insulation nonconforming or not as specified; not adhered laminates as specified; evidence of use of binders, or presence of shot.
105	Insulation not furnished in rolls (when applicable); length and width of insulation not as specified.
106	Thickness of insulation not within the specified minimum and maximum tolerances.
107	Evidence of inferior workmanship in the finished product.
Minor:	
201	Packaging, packing and boxing not as specified.
202	Gross weight of containers exceeds the specified maximum.

4.3.2 Quality conformance tests. The samples selected in accordance with 4.2.2.2 shall be subjected to the tests specified in 4.4 to determine conformance with this specification except that the thermal conductivity test (see 4.4.3) shall be conducted only when specified (see 6.2).

4.3.2.1 Action in case of failure. If any of the samples tested is found to be not in conformance with this specification, this shall be cause for rejection of the lot represented by the sample.

4.4 Test procedures.

4.4.1 Thickness. The test specimen shall be ruled off into approximately 20 square and equal areas, and the thickness measurement taken at the center of ten areas, no two of which shall have a common side. In determining the thicknesses, the test specimen shall be placed on a hard flat surface, and the penetrating pin of the depth gage shall be forced downward through the specimen, perpendicular to the flat surface as shown on figure 1. If necessary to prevent compression of the specimen by the depth-gage pin, the specimen shall first be pierced. When the point of the pin touches the flat surface, the sliding disk shall be lowered to the point of contact with the top surface of the specimen. The gage shall be withdrawn, and the distance from the point of the pin to the sliding disk shall be measured to the nearest 1/32-inch. The average of the 20 thickness measurements shall be taken as the thickness of the test specimen.

4.4.2 Weight. Each sample shall be weighed on a suitable scale to verify compliance with the requirements of 3.2.1.3 and 3.2.2.3.

4.4.3 Thermal conductivity. Thermal conductivity shall be determined in accordance with C-177.

4.4.4 Fineness of fibers. Dia. of fibers shall be determined microscopically on the basis of at least seven checks on each of the samples. The average dia. for purposes of determining conformance with 3.4 shall be the average of all measurements on all samples.

4.4.4.1 Length of fibers. A minimum of seven checks for fiber length shall be made on each of the samples. Ninety percent of the fibers selected shall be over 1/4-inch in length.

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4.4.5 Alkalinity. Weigh a 5 ± 0.01 gram (g.) representative sample of the felt, and place in a 500 milliliter (ml.) pyrex Erlenmeyer flask. Wet with 5 ml. of 95 percent ethyl alcohol and add 400 ml. of distilled water. Reflux for 4 hours ± 5 minutes. At the end of this period, disconnect the condenser and filter at once through a No. 41 Whatman paper, or its equivalent, supported in a Buechner funnel and connected to a suction source. Wash the flask and residual material three times with 25 ml. portions of hot distilled water. Titrate the combined filtrate and wash solution immediately with 0.02N H_2SO_4 using 6 to 8 drops of a 1 percent solution of phenol-red indicator, to the disappearance of the pink color. Run a blank determination on the total amount of distilled water and alcohol and substitute the titration value in the formula below:

$$\text{Percent alkalinity as } Na_2O = \frac{(A-B)N \times 0.031 \times 100}{W}$$

where A = ml H_2SO_4 required to titrate sample
B = ml H_2SO_4 required to titrate the blank
N = Normality of the H_2SO_4
W = weight of sample in grams

4.4.6 Fusing temperature. Weigh 1 g. of glass fiber into a crucible and place in a muffle furnace at room temperature. Turn all heating elements on at start of test and adjust so that the specified temperature of 1300°F is reached in 45 min. When this temperature is reached, remove crucible from furnace immediately, allow to cool, and examine visually for fusion. Fusion shall be said to have taken place if any part of the sample has melted and formed a homogeneous mass.

4.4.7 Stability. Samples of felt 4 inches square shall be encased in a metal wire screen and placed on a rack above the water level in a steam digester. The samples shall be subjected to saturated steam at 225 pounds per square inch gage (psig) for 16 hours. Samples shall then be removed and the insulation and glass fibers examined for deterioration.

4.4.8 Tensile strength,(type II only). Tensile strength shall be determined by the grab method described in method 5100 of CCC-T-191 with the following modifications: Test specimens shall be 12 by 14 inches in size. The specimens shall be clamped at the top and bottom sections by 1/2-inch pipe covered with 1/4-inch thick sponge rubber to prevent cutting and slipping of the specimens. The clamped specimens, having a test area of one sq. ft. shall be attached to the grips of the testing machine which shall separate at a rate of 2 inches per min. until rupture occurs. Tensile strength determinations shall be made on specimens before and after being subjected to soaking and heating to 1200°F. for 6 hrs. The tensile strength shall be expressed in pounds per square in. (psi) of cross-sectional area.

4.5 Possible test failures. Possible test failures are defined as follows:

(a) Weight (types I and II):
  (1) Weight of insulation not within the specified minimum (min.) and maximum (max.) values.
(b) Thermal conductivity (type I and II):
  (1) Thermal conductivity exceeds the allowable max. when tested as specified.
(c) Fineness of fibers (types I and II):
  (1) Dia. of any one fiber exceeds the specified max. average diameter of the individual fibers not within the specified min. and max. values when measured as specified.
(d) Alkalinity test (types I and II):
  (1) Alkalinity of the finished material expressed as sodium oxide (Na_2O) exceeds the max. percentage allowable.
(e) Fusing temperatures (types I and II):
  (1) Fusing temperature of the fibers less than the allowable min. when testing as specified.
(f) Stability test (types I and II):
  (1) Evidence of physical changes in the insulation and glass fibers when subjected to the required stability test.
(g) Tensile strength (type II only):
  (1) Tensile strength, expressed in pounds per square inch (psi) of cross-sectional area, less than the specified minimum when tested as specified.

4.6 Inspection of preparation for delivery. Sample rolls and shipping containers shall be selected and examined in accordance with MIL-P-116 to verify conformance to the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

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5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Rolls shall be individually covered with standard 7½-ounce (weight) burlap tubing drawn together at each end with wire ties.

5.1.2 Level C. Rolls shall be preserved and packaged in such manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity for immediate use. This may conform to the supplier's commercial practice, when such meets the requirements of this level.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Type I.

5.2.1.1 Level A. Rolls shall be packed in wood-cleated plywood, nailed wood or wire-bound boxes conforming to PPP-B-601 (overseas type), PPP-B-621 (class 2), or PPP-B-585 (class 3), respectively, at the option of the contractor. Boxes shall be closed, banded or strapped in accordance with the applicable box specification or appendix thereto. Flat steel strapping, when used, shall be type I, class B. The gross weight of the containers shall not exceed 200 pounds.

5.2.1.2 Level B. Rolls packaged as specified in 5.1.1 need not be overpacked for shipment. Alternatively, rolls may be packed in containers specified for level A (see 5.2.1) except containers shall be of the domestic type or class as applicable. Closure of boxes, when used as shipping containers, shall be as specified in the applicable box specification or appendix thereto.

5.2.1.3 Level C. Rolls shall be prepared for shipment in a manner which will insure acceptance by common carrier and safe delivery at destination. Containers or method of shipment shall comply with the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation.

5.2.2 Type II. Type II material shall require no overpacking.

5.3 Marking. In addition to any special marking required by the contract or order, packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES.

6.1 Intended use. This material is intended as thermal insulation of machinery and equipment, such as steam turbines, boilers, boiler feed pumps, and piping at temperatures up to 1200°F.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Types required (see 1.2).
(c) Length and width when other than specified (see 3.2.1 and 3.2.2).
(d) Thickness of insulation required (see 3.2.1.3 and 3.2.2.3).
(e) When thermal conductivity tests should be conducted (see 4.3.2).
(f) Levels of preservation, packaging, and packing required (see 5.1 and 5.2).
(g) Special marking, if required (see 5.3).

6.3 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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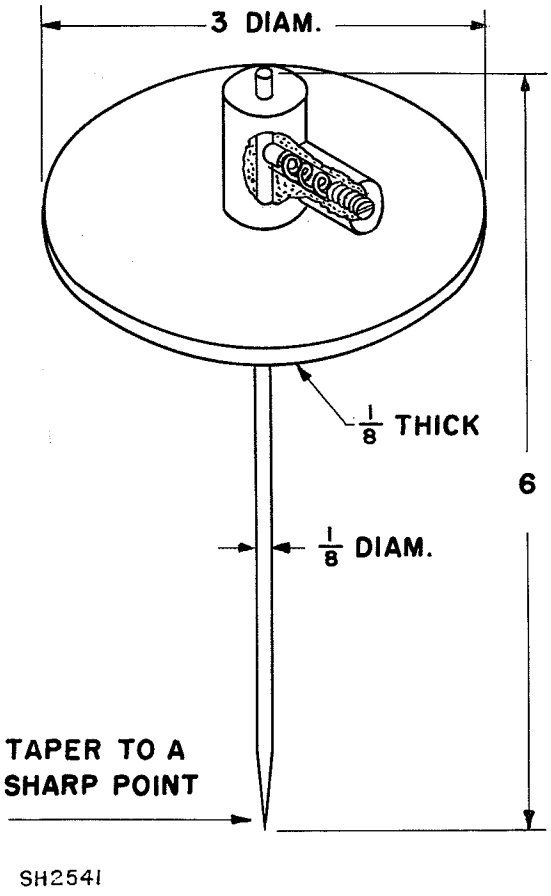


Figure 1. Depth gage for thickness measurements.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE
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